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INFORMATION HANDLING SYSTEM TO EXTEND AND SCALE A PLATFORM TELEMETRY FRAMEWORK TO A REMOTE COMPUTE DEVICE

Abstract

An information handling system includes a memory to store a system trigger event. A processor executes first and second applications, and provides data to the first application via a first data lane and to the second application via a second data lane. The processor retrieves the system trigger event from the memory and determines whether the system trigger event has been detected. In response to the system trigger event being detected, the processor transfers execution of the first application from the information handling system to a remote compute device and creates a third data lane from the data to the first application being executed on the remote compute device. The processor provides the data to the first application being executed on the remote compute device.

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Background/Summary

FIELD OF THE DISCLOSURE

[0001] The present disclosure generally relates to information handling systems, and more particularly relates to performing and scaling platform telemetry framework to a remote compute device.

BACKGROUND

[0002] As the value and use of information continues to increase, individuals and businesses seek additional ways to process and store information. One option is an information handling system. An information handling system generally processes, compiles, stores, or communicates information or data for business, personal, or other purposes. Technology and information handling needs and requirements can vary between different applications. Thus, information handling systems can also vary regarding what information is handled, how the information is handled, how much information is processed, stored, or communicated, and how quickly and efficiently the information can be processed, stored, or communicated. The variations in information handling systems allow information handling systems to be general or configured for a specific user or specific use such as financial transaction processing, airline reservations, enterprise data storage, or global communications. In addition, information handling systems can include a variety of hardware and software resources that can be configured to process, store, and communicate information and can include one or more computer systems, graphics interface systems, data storage systems, networking systems, and mobile communication systems. Information handling systems can also implement various virtualized architectures. Data and voice communications among information handling systems may be via networks that are wired, wireless, or some combination.

SUMMARY

[0003] An information handling system includes a memory to store a system trigger event. A processor may execute first and second applications. The processor may provide data to the first application via a first data lane and to the second application via a second data lane. The processor may retrieve the system trigger event from the memory and determine whether the system trigger event has been detected. In response to the system trigger event being detected, the processor may transfer execution of the first application from the information handling system to a remote compute device and create a third data lane from the data to the first application being executed on the remote compute device. The processor may provide the data to the first application being executed on the remote compute device.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0004] It will be appreciated that for simplicity and clarity of illustration, elements illustrated in the Figures are not necessarily drawn to scale. For example, the dimensions of some elements may be exaggerated relative to other elements. Embodiments incorporating teachings of the present disclosure are shown and described with respect to the drawings herein, in which:

[0005] FIG. 1 is a block diagram of a portion of a system including an information handling system and a remote compute device according to at least one embodiment of the present disclosure;

[0006] FIG. 2 is a flow diagram of a method for performing and scaling platform telemetry framework to a remote compute device according to at least one embodiment of the present disclosure; and

[0007] FIG. 3 is a block diagram of a general information handling system according to an embodiment of the present disclosure.

[0008] The use of the same reference symbols in different drawings indicates similar or identical

items.

DETAILED DESCRIPTION OF THE DRAWINGS

[0009] The following description in combination with the Figures is provided to assist in understanding the teachings disclosed herein. The description is focused on specific implementations and embodiments of the teachings and is provided to assist in describing the teachings. This focus should not be interpreted as a limitation on the scope or applicability of the teachings.

[0010] FIG. 1 illustrates a system **100** including an information handling system **102** and a remote compute device **104** according to at least one embodiment of the present disclosure. For purposes of this disclosure, an information handling system can include any instrumentality or aggregate of instrumentalities operable to compute, calculate, determine, classify, process, transmit, receive, retrieve, originate, switch, store, display, communicate, manifest, detect, record, reproduce, handle, or utilize any form of information, intelligence, or data for business, scientific, control, or other purposes. For example, an information handling system may be a personal computer (such as a desktop or laptop), tablet computer, mobile device (such as a personal digital assistant (PDA) or smart phone), server (such as a blade server or rack server), a network storage device, or any other suitable device and may vary in size, shape, performance, functionality, and price. The information handling system may include random access memory (RAM), one or more processing resources such as a central processing unit (CPU) or hardware or software control logic, ROM, and/or other types of nonvolatile memory. Additional components of the information handling system may include one or more disk drives, one or more network ports for communicating with external devices as well as various input and output (I/O) devices, such as a keyboard, a mouse, touchscreen and/or a video display. The information handling system may also include one or more buses operable to transmit communications between the various hardware components.

[0011] Information handling system **102** includes a processor **110**, multiple applications **112** and **114** to be executed by the processor or any other suitable processing component, and a memory **116**. While two applications **112** and **114** are illustrated in FIG. 1, information handling system **102** may include any suitable number of applications to be executed within the information handling system. Processor **102** may execute a telemetry service **120** and a remote/local manager **122**. In an example, remote/local manager **122** may be a part of telemetry service **120**. Application **112** may include a client portion **130** and a server portion **132**. Similarly, application **114** may include a client portion **140** and a server portion **142**. Memory **116** may store data associated with information handling system **102** including, but not limited to, one or more system trigger events **144** and one or more policy events **146**. Applications **112** and **114** may utilize and perform operations on data within information handling system **102**, such as hardware data **150** and behavior data **152**. Processor **120** may execute a data orchestration module **154** to determine whether operation of an application should be performed within information handling system **102** or remote compute device **104** as will be described herein.

[0012] Remote compute device **104** includes a processor **160** and an application **162** to be executed by the processor or any other suitable processing component. In certain examples, execution of one or more of applications **112** and **114** may be transferred from information handling system **102** to remote compute device **104** as will be described herein. While only a single application **162** is illustrated in FIG. 1, remote compute device **104** may include any suitable number of applications to be executed by processor **160** or other processing components within the remote compute device. Processor **160** may also execute a telemetry service **170** and a remote/local manager **172**. In an example, remote/local manager **172** may be a part of telemetry service **170**. Application **162** may include a client portion **190** and a server portion **192**. Application **162** may utilize and perform operations on data within remote compute device **104**, such as hardware data **180** and behavior data **182**. Processor **160** may execute a data orchestration module **184**.

[0013] In an example, remote compute device **104** may be any suitable device external to

information handling system **102**, such as an edge compute device, a dedicated compute server, a remote cloud server, or the like. In certain examples, system **100** may include any suitable number of information handling systems, such as information handling systems **103** and **105** illustrated in FIG. **1**, and any suitable number of remote compute devices, such as remote compute devices **106** and **108** illustrated in FIG. **1**, without varying from the scope of this disclosure. Information handling system **102** and remote compute device **104** may include additional components without varying from the scope of this disclosure.

[0014] During operation of information handling system **102**, applications **112** and **114** may utilize hardware data **150** and behavior data **152** of the information handling system to perform different operations. For example, applications **112** and **114** may be characterized as intelligent software applications that perform complex tasks, make predictions, automate operations, or the like. In an example, applications **112** and **114** may utilize hardware data **150** and behavior data **152** to make decisions during the performance of any one of the operations stated above. In certain examples, processor **110** may execute telemetry service **120** to provide hardware data **150** and behavior data **152** to applications **112** and **114**. Each of applications **112** and **114** may be a single entity or act in a client/server approach. For example, server portion **132** of application **112** may gather hardware data **150** and behavior data **152** via telemetry services **120** and provide the gathered data to client portion **130**. Similarly, server portion **142** of application **114** may gather hardware data **150** and behavior data **152** via telemetry services **120** and provide the gathered data to client portion **140**.

[0015] In certain examples, server portion **132** may perform different data analytics on hardware data **150** and behavior data **152**. Client portion **130** of application **112** may serve as a user experience provider for a user of information handling system **102** and perform operations on hardware data **150** and behavior data **152**. Similarly, server portion **142** may perform different data analytics on hardware data **150** and behavior data **152**. Client portion **140** of application **114** may serve as a user experience provider for a user of information handling system **102** and perform operations on hardware data **150** and behavior data **152**.

[0016] In an example, system **100** may be any suitable system, such as an information technology (IT) infrastructure, which may utilize a combination of endpoints, such as multiple information handling systems **102**, and edge resources, such as multiple remote compute devices **104**, to execute different workloads. In an example, the workloads, such as server portion **142** of application **114**, may dynamically transition from information handling system **102** to remote compute device **104** or from the remote compute device to the information handling system to effectively optimize the system resources. The workload transitions may be based on any suitable factors, such as system health, latency, workload context, or the like. In previous information handling systems, the information handling systems and the remote compute devices may require separate data orchestration modules to route data between telemetry services and the application workloads. In these previous information handling systems, the transfer of data between the telemetry services and application workloads were not scalable and added both latency and complications.

[0017] Information handling system **102** and remote compute device **104** may be improved based on operations of processor **110** executing telemetry service **120** and processor **160** executing telemetry service **170**. In particular, information handling system **102** may be improved by the transition of the execution of an application, such as application **114**, from the information handling system to remote compute device **104**. For example, processor **110**, via coordination between remote/local manager **122** of telemetry service **120** and orchestration module **154**, may automatically transfer the execution of application **112** or **114** to remote compute device **104** and re-route the corresponding data lane between hardware data **150** and behavior data **152** through telemetry service **170** of the remote compute device. In an example, information handling system **102** may be further improved by processor, via telemetry service **120**, either transferring data to application **114** locally via a data lane **158** or remotely via data lane **194**. In certain examples, the

data lane transition from data lane **158** to data lane **194** may be transparent to application **114**. [0018] During operation of information handling system **102**, any suitable number of applications, such as applications **112** and **114**, may run on the information handling system. For example, processor **110** or another processing component may execute applications **112** and **114**. In an example, processor **110**, via telemetry service **120**, may provide hardware data **150** and behavior data **152** to both applications **112** and **114**. For example, telemetry service **120** may provide hardware data **150** and behavior data **152** to application **112** via data lane **156** and to application **114** via data lane **158**. The operations of telemetry service **120** providing hardware data **150** and behavior data **152** to applications **112** and **114** being executed on information handling system **102** may be referred to herein as local data gathering operations. Thus, local gathering operations are performed when telemetry service **120** provides data from a source that is located on the same information handling system as the application utilizing the data. In certain examples, applications **112** and **114** may be instantiated on remote compute device **104** as illustrated in FIG. 1.

[0019] During operation of remote compute device **104**, any suitable number of applications, such as application **162**, may run on the remote compute device. For example, processor **160** or another processing component may execute application **162**. In an example, processor **160**, via telemetry service **170**, may provide hardware data **180** and behavior data **182** to application **162** via data lane **186**. In an example, server portion **192** may perform different data analytics on hardware data **180** and behavior data **182**. Client portion **190** of application **162** may serve as a user experience provider for a user of remote compute device **104** and perform operations on hardware data **150** and behavior data **152**.

[0020] In an example processor **110**, via data orchestration module **154**, may detect system trigger event **144**, policy event **146**, or the like. In an example processor **110**, via data orchestration module **154**, may retrieve system trigger event **144** and policy **146** from memory **116** so that the processor may monitor for the events. System trigger event **144** or policy event **146** may be any suitable event to cause the execution of server portion **132** in application **112** or server portion **142** in application **114** to be transferred to one of the remote compute devices **104**, **106**, and **108** in system **100**, such as remote compute device **104**. In an example system trigger event **144** may include, but is not limited to, a system health level, a detected latency of the data transfer, and a workload context. For example if the system health of information handling system **102** drops below a threshold level, processor **110**, via data orchestration module **154**, may determine that execution of application **114** should be transferred to remote compute device **104**. In an example, if a workload on processor **110** is above a threshold level, processor **110**, via data orchestration module **154**, may determine that execution of application **114** should be transferred to remote compute device **104**.

[0021] Based on this determination, remote/local manager **122** of telemetry service **120** may communicate with remote/local manager **172** of telemetry service **170** to perform the transfer of the execution of application **114** from information handling system **102** to remote compute device **104**. The transition or transfer of execution of server portion **142** of application **114** from information handling system **102** to remote compute device **104** is illustrated in FIG. 1 by dashed arrow between the information handling system and the remote compute device. In an example, the communication between information handling system **102** and remote compute device **104** may be over any suitable network communication interface. In certain examples, the current state of server portion **142** of application **114** may be provided to processor **160**, which in turn may initialize application **114** for execution on remote compute device **104**.

[0022] In an example, application **114** may be a containerized application. In this example and based on the determination to transfer application **114** to remote compute device **104**, remote/local manager **122** of telemetry service **120** may communicate with remote/local manager **172** of telemetry service **170** to perform the transfer of the execution of application **114** from information handling system **102** to remote compute device **104**. The transition or transfer of execution of

server portion **142** of application **114** from information handling system **102** to remote compute device **104** is illustrated in FIG. **1** by dashed arrow between the information handling system and the remote compute device. In an example, the communication between information handling system **102** and remote compute device **104** may be over any suitable network communication interface. In certain examples, the current state of the entire containerized application **114** may be provided to processor **160**, which in turn may initialize application **114** for execution on remote compute device **104**.

[0023] After the execution of application **114** is transferred to remote compute device **104**, telemetry services **170** and **120** may communicate over a network to automatically create data lane **194**. In an example, data lane **194** may be utilized to provide hardware data **150** and behavior data **152** on information handling system **102** to application **114** on remote compute device **104**. In certain examples, data lane **194** may represent a remote data operation, such that telemetry service **170** may automatically gather hardware data **150** and behavior data **152** from information handling system **102**. In certain examples, local/remote managers **122** and **172** may establish data lane **194** to ensure that telemetry service **170** gathers hardware data **150** and behavior data **152** from the correct end-point device, such as information handling system **102**.

[0024] In an example, orchestration module **154** may monitor the conditions of information handling system **102** to determine whether application **114** may no longer needed to be executed on remote compute device **104**. Similarly, orchestration module **184** may monitor the conditions of remote compute device **104** to determine whether application **114** may no longer be executed on the remote compute device. Based on the determination by based on processor **110**, via orchestration module **154**, or processor **160**, via orchestration module **154**, the execution of application **114** may be transferred back to information handling system **102**.

[0025] In certain examples, remote/local manager **122** of telemetry service **120** may communicate with remote/local manager **172** of telemetry service **170** to perform the transfer of the execution of application **114** from remote compute device **104** to information handling system **102**. After the execution of application **114** is transferred back to information handling system **102**, telemetry service **120** may provide hardware data **150** and behavior data **152** to service portion **142** of application **114** via data lane **158**. In certain examples, the current state of server portion **142** of application **114** may be provided to processor **110**, which in turn may initialize application **114** for execution again on information handling system **102**. In examples that application is a containerized application, the current state of the entire containerized application **114** may be provided to processor **110**, which in turn may initialize application **114** for execution again on information handling system **102**.

[0026] FIG. **2** is a flow diagram of a method **200** for performing and scaling platform telemetry framework to a remote compute device according to at least one embodiment of the present disclosure, starting at block **202**. It will be readily appreciated that not every method step set forth in this flow diagram is always necessary, and that certain steps of the methods may be combined, performed simultaneously, in a different order, or perhaps omitted, without varying from the scope of the disclosure. FIG. **2** may be employed in whole, or in part, processor **110** of information handling system **102** and processor **160** of remote device **104** of FIG. **1**, or any other type of controller, device, module, processor, or any combination thereof, operable to employ all, or portions of, the method of FIG. **2**.

[0027] At block **204**, one or more applications are executed in an information handling system. In an example, the applications may be characterized as intelligent software applications that perform complex tasks, make predictions, automate operations, or the like. At block **206**, data is provided to the applications. In certain examples, the data may be provided by a telemetry service of a processor of the information handling system. The data may be any suitable data of the information handling system, such as hardware data, behavior data, or the like. In an example, the data may be provided to the applications via one or more data lanes created by the telemetry service.

[0028] At block **208**, a determination is made whether a trigger event has been detected. In an example, system trigger event or policy event may be any suitable event to cause the execution of a server portion in an application to be transferred to a remote compute device. In certain examples, the system trigger event may include, but is not limited to, a system health level, a detected latency of the data transfer, and a workload context. If the trigger event has not been detected, the flow continues as described above with respect to block **206**.

[0029] If the trigger event has been detected, the execution of the application is transferred to a remote compute device at block **210**. In an example, a remote/local manager of a telemetry service in the information handling system may communicate with a remote/local manager of a telemetry service a remote compute device to perform the transfer of the execution of application from information handling system to remote compute device. In certain examples, the current state of the server portion of the application may be provided to a processor of the remote compute device, which in turn may initialize the application for execution on remote compute device. In an example, the application may be a containerized application. In this example, the current state of the entire containerized application may be provided to a processor of the remote compute device, which in turn may initialize the application for execution on remote compute device.

[0030] At block **212**, a data lane for the application is routed to the remote compute device. In an example, the telemetry services of the information handling system and the remote compute device may communicate over a network to automatically create the data lane. In an example, the data lane may be utilized to provide the hardware data and behavior data on the information handling system to the application being executed on the remote compute device. In certain examples, the local/remote managers in the information handling system and the remote compute device may establish the data lane to ensure that the telemetry service of the remote compute device gathers the hardware data and behavior data from the correct end-point device. At block **214**, the data is provided to the application on the remote compute device and the flow ends at block **216**.

[0031] FIG. **3** shows a generalized embodiment of an information handling system **300** according to an embodiment of the present disclosure. Information handling system **300** may be substantially similar to information handling system **102** of FIG. **1**. For purpose of this disclosure an information handling system can include any instrumentality or aggregate of instrumentalities operable to compute, classify, process, transmit, receive, retrieve, originate, switch, store, display, manifest, detect, record, reproduce, handle, or utilize any form of information, intelligence, or data for business, scientific, control, entertainment, or other purposes. For example, information handling system **300** can be a personal computer, a laptop computer, a smart phone, a tablet device or other consumer electronic device, a network server, a network storage device, a switch router or other network communication device, or any other suitable device and may vary in size, shape, performance, functionality, and price. Further, information handling system **300** can include processing resources for executing machine-executable code, such as a central processing unit (CPU), a programmable logic array (PLA), an embedded device such as a System-on-a-Chip (SoC), or other control logic hardware. Information handling system **300** can also include one or more computer-readable medium for storing machine-executable code, such as software or data. Additional components of information handling system **300** can include one or more storage devices that can store machine-executable code, one or more communications ports for communicating with external devices, and various input and output (I/O) devices, such as a keyboard, a mouse, and a video display. Information handling system **300** can also include one or more buses operable to transmit information between the various hardware components.

[0032] Information handling system **300** can include devices or modules that embody one or more of the devices or modules described below and operates to perform one or more of the methods described below. Information handling system **300** includes a processors **302** and **304**, an input/output (I/O) interface **310**, memories **320** and **325**, a graphics interface **330**, a basic input and output system/universal extensible firmware interface (BIOS/UEFI) module **340**, a disk controller

350, a hard disk drive (HDD) **354**, an optical disk drive (ODD) **356**, a disk emulator **360** connected to an external solid state drive (SSD) **362**, an I/O bridge **370**, one or more add-on resources **374**, a trusted platform module (TPM) **376**, a network interface **380**, a management device **390**, and a power supply **395**. Processors **302** and **304**, I/O interface **310**, memory **320**, graphics interface **330**, BIOS/UEFI module **340**, disk controller **350**, HDD **354**, ODD **356**, disk emulator **360**, SSD **362**, I/O bridge **370**, add-on resources **374**, TPM **376**, and network interface **380** operate together to provide a host environment of information handling system **300** that operates to provide the data processing functionality of the information handling system. The host environment operates to execute machine-executable code, including platform BIOS/UEFI code, device firmware, operating system code, applications, programs, and the like, to perform the data processing tasks associated with information handling system **300**.

[0033] In the host environment, processor **302** is connected to I/O interface **310** via processor interface **306**, and processor **304** is connected to the I/O interface via processor interface **308**. Memory **320** is connected to processor **302** via a memory interface **322**. Memory **325** is connected to processor **304** via a memory interface **327**. Graphics interface **330** is connected to I/O interface **310** via a graphics interface **332** and provides a video display output **336** to a video display **334**. In a particular embodiment, information handling system **300** includes separate memories that are dedicated to each of processors **302** and **304** via separate memory interfaces. An example of memories **320** and **330** include random access memory (RAM) such as static RAM (SRAM), dynamic RAM (DRAM), non-volatile RAM (NV-RAM), or the like, read only memory (ROM), another type of memory, or a combination thereof.

[0034] BIOS/UEFI module **340**, disk controller **350**, and I/O bridge **370** are connected to I/O interface **310** via an I/O channel **312**. An example of I/O channel **312** includes a Peripheral Component Interconnect (PCI) interface, a PCI-Extended (PCI-X) interface, a high-speed PCI-Express (PCIe) interface, another industry standard or proprietary communication interface, or a combination thereof. I/O interface **310** can also include one or more other I/O interfaces, including an Industry Standard Architecture (ISA) interface, a Small Computer Serial Interface (SCSI) interface, an Inter-Integrated Circuit (I2C) interface, a System Packet Interface (SPI), a Universal Serial Bus (USB), another interface, or a combination thereof. BIOS/UEFI module **340** includes BIOS/UEFI code operable to detect resources within information handling system **300**, to provide drivers for the resources, initialize the resources, and access the resources. BIOS/UEFI module **340** includes code that operates to detect resources within information handling system **300**, to provide drivers for the resources, to initialize the resources, and to access the resources.

[0035] Disk controller **350** includes a disk interface **352** that connects the disk controller to HDD **354**, to ODD **356**, and to disk emulator **360**. An example of disk interface **352** includes an Integrated Drive Electronics (IDE) interface, an Advanced Technology Attachment (ATA) such as a parallel ATA (PATA) interface or a serial ATA (SATA) interface, a SCSI interface, a USB interface, a proprietary interface, or a combination thereof. Disk emulator **360** permits SSD **364** to be connected to information handling system **300** via an external interface **362**. An example of external interface **362** includes a USB interface, an IEEE 4394 (Firewire) interface, a proprietary interface, or a combination thereof. Alternatively, solid-state drive **364** can be disposed within information handling system **300**.

[0036] I/O bridge **370** includes a peripheral interface **372** that connects the I/O bridge to add-on resource **374**, to TPM **376**, and to network interface **380**. Peripheral interface **372** can be the same type of interface as I/O channel **312** or can be a different type of interface. As such, I/O bridge **370** extends the capacity of I/O channel **312** when peripheral interface **372** and the I/O channel are of the same type, and the I/O bridge translates information from a format suitable to the I/O channel to a format suitable to the peripheral channel **372** when they are of a different type. Add-on resource **374** can include a data storage system, an additional graphics interface, a network interface card (NIC), a sound/video processing card, another add-on resource, or a combination thereof. Add-on

resource **374** can be on a main circuit board, on separate circuit board or add-in card disposed within information handling system **300**, a device that is external to the information handling system, or a combination thereof.

[0037] Network interface **380** represents a NIC disposed within information handling system **300**, on a main circuit board of the information handling system, integrated onto another component such as I/O interface **310**, in another suitable location, or a combination thereof. Network interface device **380** includes network channels **382** and **384** that provide interfaces to devices that are external to information handling system **300**. In a particular embodiment, network channels **382** and **384** are of a different type than peripheral channel **372** and network interface **380** translates information from a format suitable to the peripheral channel to a format suitable to external devices. An example of network channels **382** and **384** includes InfiniBand channels, Fibre Channel channels, Gigabit Ethernet channels, proprietary channel architectures, or a combination thereof. Network channels **382** and **384** can be connected to external network resources (not illustrated). The network resource can include another information handling system, a data storage system, another network, a grid management system, another suitable resource, or a combination thereof.

[0038] Management device **390** represents one or more processing devices, such as a dedicated baseboard management controller (BMC) System-on-a-Chip (SoC) device, one or more associated memory devices, one or more network interface devices, a complex programmable logic device (CPLD), and the like, which operate together to provide the management environment for information handling system **300**. In particular, management device **390** is connected to various components of the host environment via various internal communication interfaces, such as a Low Pin Count (LPC) interface, an Inter-Integrated-Circuit (I2C) interface, a PCIe interface, or the like, to provide an out-of-band (OOB) mechanism to retrieve information related to the operation of the host environment, to provide BIOS/UEFI or system firmware updates, to manage non-processing components of information handling system **300**, such as system cooling fans and power supplies. Management device **390** can include a network connection to an external management system, and the management device can communicate with the management system to report status information for information handling system **300**, to receive BIOS/UEFI or system firmware updates, or to perform other task for managing and controlling the operation of information handling system **300**.

[0039] Management device **390** can operate off of a separate power plane from the components of the host environment so that the management device receives power to manage information handling system **300** when the information handling system is otherwise shut down. An example of management device **390** include a commercially available BMC product or other device that operates in accordance with an Intelligent Platform Management Initiative (IPMI) specification, a Web Services Management (WSMan) interface, a Redfish Application Programming Interface (API), another Distributed Management Task Force (DMTF), or other management standard, and can include an Integrated Dell Remote Access Controller (iDRAC), an Embedded Controller (EC), or the like. Management device **390** may further include associated memory devices, logic devices, security devices, or the like, as needed, or desired.

[0040] Although only a few exemplary embodiments have been described in detail herein, those skilled in the art will readily appreciate that many modifications are possible in the exemplary embodiments without materially departing from the novel teachings and advantages of the embodiments of the present disclosure. Accordingly, all such modifications are intended to be included within the scope of the embodiments of the present disclosure as defined in the following claims. In the claims, means-plus-function clauses are intended to cover the structures described herein as performing the recited function and not only structural equivalents, but also equivalent structures.

Claims

- 1.** An information handling system comprising: a memory to store a plurality of system trigger events; and a processor to communicate with the memory, the processor to: execute first and second applications; provide data to the first application via a first data lane and to the second application via a second data lane; retrieve the system trigger events from the memory; determine whether one of the system trigger events has been detected; and in response to the one of the system trigger events being detected, the processor further to: transfer execution of the first application from the information handling system to a remote compute device; create a third data lane from the data to the first application being executed on the remote compute device; and provide the data to the first application being executed on the remote compute device over the third data lane.
- 2.** The information handling system of claim 1, wherein the processor further to: determine whether a condition exists to cause the execution of the first application to be transferred back to the information handling; and in response to the condition existing, transfer the execution of the first application to the information handling system.
- 3.** The information handling system of claim 2, wherein the processor further to: re-establish the first data lane between the data and the first application being executed on the information handling system.
- 4.** The information handling system of claim 1, wherein the first application includes a client portion and a server portion.
- 5.** The information handling system of claim 4, wherein the transfer of the execution of the first application from the information handling system to the remote compute device, the processor further to: transfer the server portion of the application to the remote compute device.
- 6.** The information handling system of claim 1, wherein the processor further to: communicate with a second processor of the remote compute device to establish the third data lane.
- 7.** The information handling system of claim 1, wherein the processor further to: execute a telemetry service to provide the data over the first data lane to the first application being executed in the information handling system.
- 8.** The information handling system of claim 1, wherein the system trigger events includes a system health level, a detected latency of the data transfer, and a workload context.
- 9.** A method comprising: storing a plurality of system trigger events in a memory of an information handling system; executing, by a processor of the information handling system, first and second applications; providing, by the processor, data to the first application via a first data lane and to the second application via a second data lane; retrieving the system trigger events from the memory; determining whether one of the system trigger events has been detected; and in response to the one of the system trigger events being detected: transferring execution of the first application from the information handling system to a remote compute device; creating a third data lane from the data to the first application being executed on the remote compute device; and providing, by the processor, the data to the first application being executed on the remote compute device over the third data lane.
- 10.** The method of claim 9, further comprising: determining whether a condition exists to cause the execution of the first application to be transferred back to the information handling; and in response to the condition existing, transferring the execution of the first application to the information handling system.
- 11.** The method of claim 10, further comprising: re-establishing the first data lane between the data and the first application being executed on the information handling system.
- 12.** The method of claim 9, wherein the first application includes a client portion and a server portion.

- 13.** The method of claim 12, wherein the transferring of the execution of the first application from the information handling system to the remote compute device, the method further comprises: transferring the server portion of the application to the remote compute device.
- 14.** The method of claim 9, further comprising: communicating with a second processor of the remote compute device to establish the third data lane.
- 15.** The method of claim 9, further comprising: executing a telemetry service to provide the data over the first data lane to the first application being executed in the information handling system.
- 16.** The method of claim 9, wherein the system trigger events includes a system health level, a detected latency of the data transfer, and a workload context.
- 17.** A non-transitory computer-readable medium to store instructions that are executable to perform operations comprising: executing first and second applications; providing data to the first application via a first data lane and to the second application via a second data lane; and if a system trigger event is detected, then: transferring execution of the first application from the information handling system to a remote compute device; creating a third data lane from the data to the first application being executed on the remote compute device; and providing the data to the first application being executed on the remote compute device over the third data lane.
- 18.** The non-transitory computer-readable medium of claim 9, the operations further comprising: determining whether a condition exists to cause the execution of the first application to be transferred back to the information handling; and in response to the condition existing, transferring the execution of the first application to the information handling system.
- 19.** The non-transitory computer-readable medium of claim 10, the operations further comprising re-establishing the first data lane between the data and the first application being executed on the information handling system.
- 20.** The non-transitory computer-readable medium of claim 9, the operations further comprising executing a telemetry service to provide the data over the first data lane to the first application being executed in the information handling system.
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